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Model-based feedback controller design for dual actuated atomic force microscopy

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ABSTRACT

In atomic force microscopy (AFM) the imaging speed is strongly limited by the bandwidth of the feedback loop that controls the interaction force between the measurement tip and the sample. A significant increase in closed-loop bandwidth without sacrificing positioning range can be achieved by combining a long-range, low-bandwidth actuator with a short-range, high-bandwidth actuator, forming a dual actuated system. This contribution discusses the design of a model-based feedback controller that controls the tip-sample force in dual actuated AFM. Special emphasis is given on guaranteeing robust stability of the feedback loop under influence of variations in the dynamical behavior of the system, and to prevent strong destructive interference between both actuators. To prevent instability of the feedback loop due to saturation of the short-range actuator, an anti-windup controller is presented that robustly stabilizes the system under all imaging conditions. The designed feedback controller is implemented on a prototype dual actuated AFM system, and demonstrates a disturbance rejection bandwidth of 20 kHz, which is about 20 times faster than the model-based controlled single actuated system. AFM images are obtained verifying a significant reduction of force variations between the tip and the sample while imaging. The faster control of the tip-sample force reduces the residual tracking error and, thus, reduces the chance of damage or wear of the tip and the sample, and allows for faster imaging.

1. Introduction

Atomic force microscopes (AFMs) are mechanical microscopes in which the sample is probed by a very sharp tip, while scanning the sample relative to the measurement tip in a raster scan pattern [1]. In AFM the measurement tip is mounted on the free end of a micro-cantilever, which allows to measure the force between the tip and sample by detecting the micro-cantilever deflection via a optical sensing system [2]. During imaging the force between the tip and the sample is controlled via a feedback loop in order to minimize the chances of damage or wear of the tip and the sample, and to convert the force measurement into a measurement of the sample topography. To provide the scanning motion and the control of the tip-sample force, most often a piezo-based positioning stage is used to allow positioning of the sample relative to the measurement tip in all three spatial directions with nanometer precision.

Due to its capability of providing (sub-) nanometer resolution images in all kinds of environmental conditions, AFM has become a very popular tool in the fields of nanotechnology, micro-biology and material sciences. One of the main limitations of AFM, however, is its relatively low imaging speed, taking in the order of several minutes per frame for most commercially available AFM-systems, which results in a low through-put of these systems, and restricts its application to the imaging of fairly static features [3].

In order to improve the imaging speed of AFM, a vast amount of research is dedicated to improve both the speed of the lateral scanning motion, as well as faster control of the tip-sample force [4,5]. Recently, various prototype AFM systems are reported in literature allowing high speed lateral scanning via improved mechanical designs of the scanning stage [6,7], as well as improved control of the scanning motion [8–12]. Some laboratory prototype AFM systems are reported with scanning speeds allowing to capture up to several frames per second [6,7,13,14]. The improved scanning speed, however, is putting strong demands on the control of the tip-sample force during imaging, which is considered as the main limitation on high-speed AFM imaging nowadays [3].

In AFM the force between the tip and the sample can be measured in both contact-mode, as well as in tapping mode [4]. Often, in high-speed imaging applications contact mode imaging is preferred due to its faster, direct measurement of the tip-sample force, although recently significant improvements also have been made on high speed tapping mode imaging [16–18]. For both imaging modes high-bandwidth control of the tip-sample force is very important in order to provide a good measurement of the sample topography, and to prevent damage or wear of the tip and sample, particulary when imaging fragile biological specimen [3]. Recently, improved control of the tip-sample force by use of modern

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model-based control techniques has been investigated [19,20], showing a significant improvement in closed-loop bandwidth as compared to the classical PI-controllers used in most commercially available AFM systems nowadays. Moreover, prototype systems are reported in literature with improved actuator designs to allow faster manipulation of the tip-sample separation, utilizing high-bandwidth piezo-based actuators [6,7,21,22], as well as MEMS-based actuators with integrated measurement probes [23,24].

Optimizing these type of actuators for high-bandwidth positioning, however, often comes at the cost of a significant reduction in positioning range, limiting the application of these actuators to samples with relatively small topographic features when used as the sole means of controlling the tip-sample force. Meanwhile, most often in AFM imaging the largest topographic variations to be tracked by the vertical feedback loop are relatively slow varying (e.g. due to the sample tilt), and can be easily tracked by a relatively low-bandwidth actuator. This aspect allows to combine a long-range, low-bandwidth actuator with a short-range, high-bandwidth actuator in order to obtain a system with both a high control bandwidth as well as a large effective positioning range. This technique is generally referred to as dual-actuation, and has been thoroughly investigated for implementations on Hard Disk Drives (HDD) (e.g. [25,26]), and also found its way towards scanning probe microscopes [22,23,27–29], showing vast improvements on the control-bandwidths of the vertical feedback loops as compared to single actuated system.

In the majority of the earlier published work on dual-actuated AFM, the main focuss it put on the development of the short-range, high-bandwidth actuators, while the design of the feedback controller is done in a more heuristic manner. The contribution of this paper is focussed on developing a systematic approach to the design of a feedback controller for a given dual-actuated AFM, utilizing modern model-based control techniques. The experimental setup used in this research is introduced in Section 2, and subsequently the overall control problem and design procedure is outlined. When designing such feedback controllers, a major limitation on the closed-loop bandwidth is often posed by the variations in dynamical behavior of the actuators, which occurs when changing the measurement probe or sample, or by variations of their alignments with the scanner [30]. In Section 3 an identification procedure is proposed in order to identify the system's dynamical behavior and its variations. This allows to analyze the achievable feedback bandwidth, as well as the robustness of the designed dual-actuated closed-loop system. The design of the model-based feedback controller is discussed in Section 4, putting special emphasis on how to prevent strong destructive interference between both actuators, and on taking into account the robustness of the controlled system. An important consideration in the design of a controller for these type of dual actuated systems is the limited actuation range of the short-range actuator. As the short-range actuator might saturate when imaging large sample features, this can lead to instability of the closed-loop system. In Section 5 an approach is discussed to analyze the stability of the feedback loop during saturation of the short-range actuator, and an anti-windup controller is introduced that assures stability of the feedback loop under all imaging conditions. The improved closed-loop bandwidth of the dual-actuated system is experimentally verified in Section 6, and demonstrated in an imaging application, enabling higher imaging speeds.

2. Problem analysis and design strategy

A dual actuated AFM is a rather complex system, and detailed understanding of the system is required in order to design a well performing controller for it. In this section the experimental setup is introduced, and subsequently the control problem and the design strategy for the controller is outlined.

2.1. Experimental setup

The system used in this research is based on a commercially available AFM-system (Multimode V, Bruker, Santa Barbara, USA), and shown schematically in Fig. 1. The force between the tip and the sample is measured by reflecting a laser beam off the free end of the micro cantilever where the measurement tip is mounted at, and measuring the deflected laser spot with a segmented photodiode. A piezoelectric tube-scanner (J-Scanner, Bruker, Santa Barbara, USA), is utilized to provide the scanning motion of the sample during imaging, as well as the vertical positioning of the sample to control the tip-sample force. This piezoelectric tube scanner will be referred to as the 'long-range actuator', and is driven by an external piezo amplifier (PZD700, Trek, Medina, USA) which combined provides a vertical positioning range of 5 μm. This system is extended to a dual actuated system by using a small piezoelectric plate actuator (CMAP12, Noliac, Kvistgaard, Denmark) which is glued onto the cantilever holder to form the new mounting spot for the cantilever-chip (cf. Fig. 1), and allows vertical positioning of the cantilever-chip. The plate-actuator is driven by a piezo amplifier (A400, FLC-electronics, Partille, Sweden), of which the output voltage is restricted to 20 V_{pp} to prevent it from running into its current limitations while dynamically driving the 36 nF capacitive load of the piezoelectric plate actuator. This allows a maximum positioning range of about 0.5 μm for the piezoelectric plate actuator, which will be referred to as the 'short-range actuator'. The resulting dual actuated AFM is a simple form of an overactuated system in which the tip-sample force can be controlled via the two actuators.

2.2. Controller design problem

To minimize the force variations between the measurement tip and the sample during imaging, a feedback controller should be designed which provides the highest possible control bandwidth given the two actuators. It should be noted that the pushing the control bandwidth of the system as high as possible can have negative influence on the accuracy of the topography measurement, as discussed in [31], and is therefore not always the best choice for AFM application where high priority is given to the

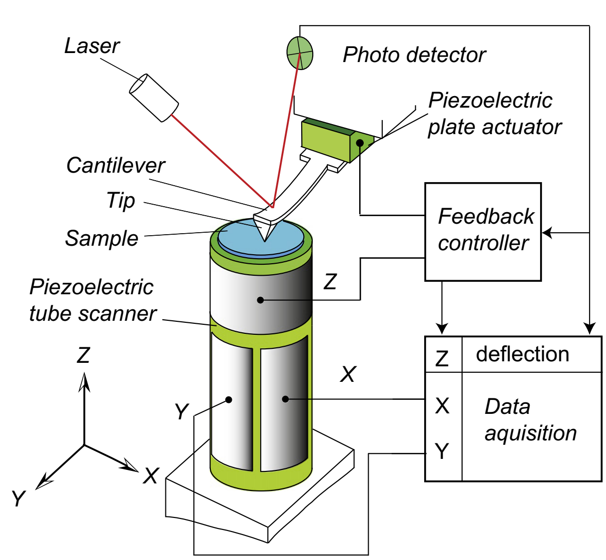

Fig. 1. Schematic representation of a dual actuated atomic force microscope.

metrological aspects of the system. However, in imaging application where minimizing the imaging forces to prevent damage to the tip and sample is most important, for instance when imaging fragile biological samples at high speed, fast control of the tip-sample force has the highest priority, which is the scope of this research.

When designing a controller for dual actuated AFM, it should be considered that during normal imaging experiments various samples and cantilevers might be used, which can cause variations in the dynamic behavior of the system. Apart from the weight variations introduced by the various measurement probes and samples, the system is also sensitive to slight variations in the alignment of the sample and cantilevers with respect to the actuators, which can cause couplings towards the lateral resonance modes of the system [30]. Although during each individual imaging experiment the dynamical behavior of the system might be fairly constant, re-identifying the system and adjusting the controller before each imaging experiment is not desirable, as this would be time-consuming, would require additional skills of the operator, and is likely to cause damage or contamination to the tip and sample. It is therefore important that the designed feedback controller provides satisfactory performance and stability for all possible combinations of samples and measurement probes and the possible variations in the alignments over various imaging experiments.

Other than for conventional, single actuated AFMs, one important design consideration for the feedback controller of a dual-actuated AFM is posed by the limited positioning range of the short-range actuator. To prevent damage to the tip and the sample, the control system should be designed such that under normal imaging conditions the chance of saturation of the short-range actuator is minimal. The underlying assumption of dual actuated AFM, is that the long-range actuator primarily tracks the larger amplitude low-frequency topography variations, while the short-range actuator has sufficient range to track high frequency topography variations that typically have smaller amplitudes. However, incidental large disturbances, for instance during the initialization phase of the imaging experiment or due to large steps in the sample surface might still drive the short-range actuator towards saturation, and the controller should guarantee overall stability and fast recovery of the system if such an event occurs. The design of the controller is therefore split into two stages; first a nominal linear feedback controller is designed aimed at optimal performance within the nominal operating regime of the system, and secondly an anti-windup controller is designed which guarantees stability and fast recovery of the system in case of saturation of the short-range actuator. The design of the nominal feedback controller constitutes the major part of the design process, and the used design strategy is further outlined in the remainder of this section. The design and implementation of the anti-windup controller is discussed in Section 5.

2.3. Design procedure nominal feedback controller

Although the dual-actuated servo systems are multi input/single output (MISO) systems, several controller design methods can be found in literature which are based on SISO controller design techniques. Examples are the nested or Master–Slave structure (e.g. [23]), and the decoupled Master–Slave structure [32], in which the individual control paths are sequentially tuned. Alternatively, in [33] a method is proposed in which first design of the controller is focussed on tuning the frequency separation between both control-paths, while in a second step the overall feedback control action is tuned. Although these SISO-based design methods provide intuitive ways to design a feedback controller for dual actuated systems, the sequential design of the different controller facets, and the fact that specific controller structures are pre-imposed might not always lead to the most optimal performance of the overall system.

Model-based controller design methods are well suited for these type of MISO systems, and have the advantage that the overall closed-loop behavior can be directly enforced, without pre-imposing a certain structure on the controller. Model-based feedback controller designs for dual actuated systems are reported in literature for dual-stage HDD drives (see [26] and the reference therein), and more recently on dual-actuated AFM [27]. As outlined above, an important design criterion for the feedback controller in AFM is to provide robustness against the dynamical variations of the system, occurring for instance when using varying cantilevers and samples and by variations of the alignment. Model-based controller design guaranteeing both robust stability as well as robust performance against dynamic uncertainties of the system are shown in [34] for dual-stage HDD drives by over-bounding the multiplicative dynamical uncertainty of the system with weighting filters and applying μ-synthesis. To reduce conservatism of this method, a tight over-bound of the dynamic uncertainty is needed. This, however, might lead to very high model orders and consequently high order feedback controllers which may be limited at the practical implementation, particulary when a high control bandwidth (and thus sampling rate) is required. In [35] the model order is significantly reduced by capturing the dynamic uncertainty in a parametric sense. For the actuators used in this particular research, however, the uncertain dynamical modes in the frequency range of interest are too numerous, and therefore it is impossible to sufficiently capture the dynamical uncertainty in a parametric sense (cf. Fig. 2).

In this work the feedback controller for the dual actuated AFM is therefore designed based on a model capturing only the nominal dynamics of the actuators, but by ensuring sufficient attenuation of the dynamical uncertainty by proper choice of the weighting filters, as discussed in Section 4. After the design of the feedback controller the robustness of the control system is verified via μ-analysis. As a first step of this design process, the actuator dynamics are identified for various combinations of measurements probes and cantilevers to analyze the dynamic behavior of the system over various measurement experiments, and to acquire a model of the systems dynamics to be used for model-based control design.

3. System identification

The frequency responses of the vertical actuators of the AFM-system can be identified by applying a reference signal to the actuators and measuring the cantilever deflection, while the tip is in contact with the sample and the scanning motion is disabled. To identify both the nominal dynamical behavior of the system, as well as its dynamical variations over various imaging experiments, the frequency responses of both actuators are measured 12 different times while varying the type and the alignment of the measurement probes and/or sample discs. The various load conditions are representative for actual AFM-measurements with the typical variation in the system setup. As the system used in this particular research typically is used for high-speed contact-mode imaging, only AFM-cantilevers with high free resonance frequency are considered, ($300\text{ kHz} < f_0 < 600\text{ kHz}$). The weight of the sample discs used ranges from 0.5 to 1 g. The frequency responses are recorded by a network analyzer (4395A, Agilent, Santa Clara, USA), and stored in vectors denoted $\Phi_{j,k}(\omega_f)$, with $\omega_f \in \Omega_f$ the frequency points ranging from 10 Hz to 1 MHz, and integer $k \in [0, n]$ denoting the measurement number. Subscript j is used to denote the corresponding actuator, with $j = 1$ being the long-range actuator, and $j = 2$ the short-range actuator. The results of

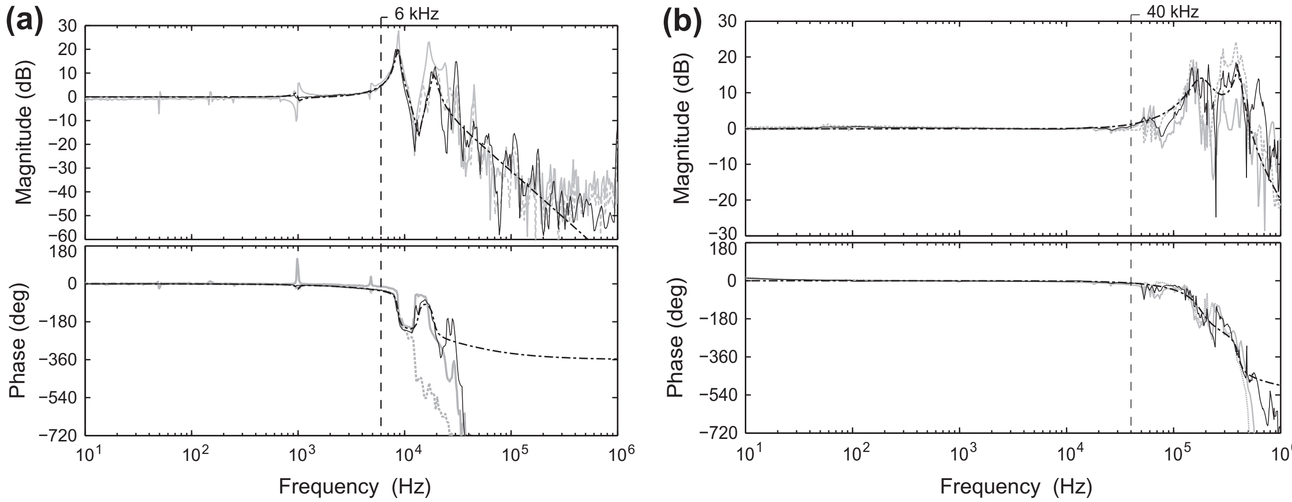

Fig. 2. Frequency responses from the long-range actuator (a), and of the short-range actuator (b), for two different measurement runs (gray, dashed and solid lines), the unparameterized nominal model (black, solid lines), and the parameterized models capturing the nominal dynamics (black, dashed-dotted lines).

two typical realizations of the identification experiments for both actuators are shown in Fig. 2 (gray, solid and dashed lines). Based on the measurement data a frequency response model can be determined that contains the dynamical behavior of the system as well as its uncertainty:

$$\mathcal{G}(\omega_f,\Delta)=\begin{cases}\Gamma_1(\omega_f)\cdot(1+\Delta_1(\omega_f)\Psi_1(\omega_f))\\ \Gamma_2(\omega_f)\cdot(1+\Delta_2(\omega_f)\Psi_2(\omega_f))\end{cases},|\Delta_1(\omega_f)|\leqslant 1,|\Delta_2(\omega_f)|\leqslant 1,\forall\omega_f\in\Omega_f, \tag{1}$$

with the 'nominal' frequency response $\Gamma_j(\omega_f)\in\mathbb{C}$ which is at the center of the set of frequency responses, and calculated as:

$$\Gamma_j(\omega_f)=\arg\min_{\Gamma_j(\omega_f)}\max_{k=1\ldots n}|\Phi_{j,k}(\omega_f)-\Gamma_j(\omega_f)|, \tag{2}$$

and with $\Psi_{j,k}(\omega_f)$ bounding the multiplicative uncertainty at each frequency point, given by:

$$\Psi_j(\omega_f)=\max_{k=1\ldots n}\left|\frac{\Phi_{j,k}(\omega_f)-\Gamma_j(\omega_f)}{\Gamma_j(\omega_f)}\right|. \tag{3}$$

Parameter $\Delta(\omega_f)\in\mathbb{C}$ is a normalized, complex uncertain parameter which describes the dynamical uncertainty of the system. Although the model of Eq. (1) might capture a larger set of frequency responses than the actual dynamics of the system, and therefore might be a bit conservative, it can provide useful information for controller design and stability analysis. Fig. 2 shows the nominal responses $\Gamma_j(s)$ for both actuators (black, solid lines). The corresponding multiplicative uncertainties $\Psi_j(\omega_f)$ are shown in Fig. 3, showing that the dynamical uncertainty of both actuators significantly increases at higher frequencies. Moreover, Fig. 3 shows that the dynamic uncertainty does not only occur around the longitudinal resonance modes, but also due to couplings towards the lateral resonance modes which are very sensitive for the variations in alignment of the sample and measurement probe with respect to the scanner (e.g. for the long-range actuator at 1 kHz, and for the short-range actuator at 40 kHz).

At frequency regions where the magnitude of the multiplicative dynamical uncertainty is larger than one ($\Psi_j(\omega_f)\geqslant 1$), the dynamical variation of the system is more then 100% of its nominal value, and hence the phase behavior of the system is fully unpredictable in that frequency region (cf. Eq. (1)). Therefore, the frequency at which the multiplicative uncertainty exceeds the 0 dB-line can be seen as the limit on the actuation bandwidth for each actuator at which stability of the system still can be guaranteed. From Fig. 3 it can therefore be deduced that for the long-range actuator the

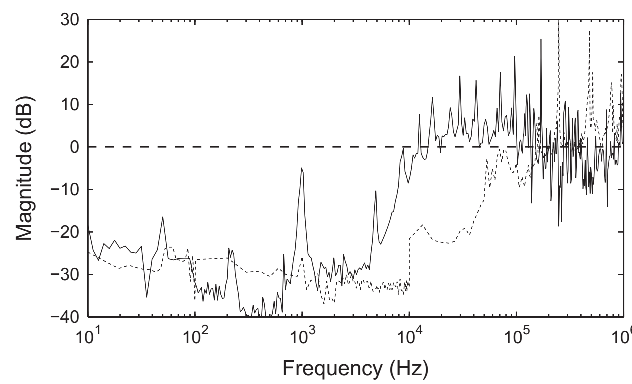

Fig. 3. Bode magnitude plot of the multiplicative dynamic uncertainty $\Psi_j(\omega_f)$ for the long-range actuator (solid) and for the short-range actuator (dashed).

maximum allowable bandwidth is about 6 kHz, and for the short-range actuator the maximum bandwidth is about 40 kHz. These bandwidth limitations are explicitly taken into account in the design of the nominal feedback controller. The robustness of the feedback controller is verified afterwards via μ-analysis, using the frequency response model of Eq. (1).

In order to allow model-based design of the feedback controller, parameterized transfer function models are fitted that are based on the nominal frequency responses $\Gamma_j(\omega_f)$ (cf. Eq. (2)) using least squares data-fitting via the `matlab`-function `fitfrd.mat`. The 7th order model $G_1(s)$ of the long-stroke actuator, and the 5th order model $G_2(s)$ of the short-range actuator are shown in Fig. 2 (black, dashed-dotted lines), and are used for controller synthesis as discussed in the next section.

4. Model-based feedback controller design

To enable model-based controller design the system is cast in the mixed-sensitivity framework as depicted in Fig. 4. The following transfer function blocks can be recognized: the nominal actuator dynamics $G(s)=diag[G_1(s),\ G_2(s)]$, $W_e(s)$ is the weighting filter on the control error, $W_y(s)=diag[W_{y1}(s),\ W_{y2}(s)]$ are weighting filters on the actuator outputs, and $W_u(s)=diag[W_{u1}(s),\ W_{u2}(s)]$ are the weighting filters on the controller outputs. The optimization objective is to find the parameters $\hat{\theta}$ of controller

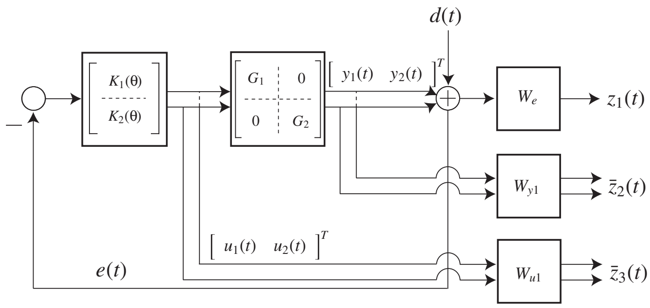

Fig. 4. Block diagram of the system representation used for controller synthesis, with controller $K(s,\theta)$, plant dynamics $G(s)$, and weighting filters $W_e(s)$, $W_y(s)$, and $W_u(s)$.

$K(s,\theta) = [K_1(s,\theta),\ K_2(s,\theta)]^T$ which minimizes the H_∞-norm of the system:

$$\gamma \geqslant \min_\theta \left\| \begin{matrix} W_e(s)\cdot S(s,\theta) \\ W_y(s)\cdot T(s,\theta) \\ W_u(s)\cdot S(s,\theta)\cdot K(s,\theta) \end{matrix} \right\|_\infty , \tag{4}$$

with $S(s,\theta) = (1 + \sum\{G(s)\cdot K(s,\theta)\})^{-1}$ the sensitivity function, and $T(s,\theta) = G(s)\cdot K(s,\theta)\cdot S(s,\theta)$ the transfer towards the individual actuator outputs.

The weighting filters are used to enforce the desired behavior on the closed-loop system. Fig. 5 shows the Bode magnitude responses of the individual (inverse) weighting filters together with the corresponding transfer function of the resulting closed-loop system. The most important criteria is the disturbance rejection, which is enforced by choosing the weighting filter $W_e(s)$ as an inverse high-pass filter (cf. Fig. 5a), pushing the bandwidth as high as possible.

As discussed in Section 3 the actuation bandwidths should be limited to 6 kHz for the long-range actuator and 40 kHz for the short-range actuators to prevent instability due to excitation of the uncertain dynamics. To this end, weighting filters $W_{u_1}^{-1}(s)$, $W_{y_1}^{-1}(s)$, and $W_{y_2}^{-1}(s)$ are chosen as first order low-pass filters with corresponding cornering frequency. The dc-gain of these filters is set to 2 in order not to put too strong weights on potential peaking above the 0 dB line of these transfer functions, that naturally occurs just before the cross-over frequency (cf. Fig. 5b–d). Weighting filter $W_{u_2}^{-1}(s)$ provides 2nd order roll-off in order to enforce a steeper roll-off of the controller output for the short-range actuator at higher frequencies (cf. Fig. 5e). To enforce the frequency separation between both actuators, a weight is put on the control action of the short-range actuator at lower frequency via weighting filter $W_{u_2}^{-1}(s)$. This frequency separation should be carefully designed to minimize the chance of saturation of the short-range actuator during imaging.

4.1. Frequency separation

One important design aspect for dual actuated systems, is that if the phase difference between both control paths gets too large in a particular frequency region, the actuators might start to 'fight' each other as in compensating part of each others motion. Fighting between both actuators should be avoided as this could result in large and inefficient controller outputs, which might in turn cause undesirable excitation of (non-linear) dynamics and would increase the chance of actuator saturation during imaging.

When actuator fighting occurs in a particular frequency region, the magnitude of the overall complementary sensitivity function (i.e. the sum of the actuator outputs) is smaller than the largest magnitude of the individual actuator outputs in that frequency region; $|T_1(i\omega) + T_2(i\omega)| < max\{|T_1(i\omega)|, |T_2(i\omega)|\}$. As depicted in Fig. 6 this destructive interference between the two actuators starts from a phase difference of 90° for $min\{|T_1(i\omega)|, |T_2(i\omega)|\} \ll max\{|T_1(i\omega)|, |T_2(i\omega)|\}$ up to 120° at the transition frequency $|T_1(i\omega)| = |T_2(i\omega)|$.

To prevent too large phase differences the transition between both actuators is restricted to a first order roll-on for the short-range actuator at lower frequency, i.e. having a 90° phase lead with respect to the long-range actuator. After the transition frequency $\omega_t|(|T_1(i\omega_t)| = |T_2(i\omega_t)|)$ the phase difference between both control paths is largely determined by the roll-off for the long-range actuator. Naturally one would like to choose the transition frequency as high as possible to make maximum use of the positioning range of the long-range actuator. However, the transition frequency should not be chosen too high, as this might require a steeper roll-off for the long-range actuator to provide sufficient attenuation of its uncertain dynamics at higher frequencies. Such rapid roll-off

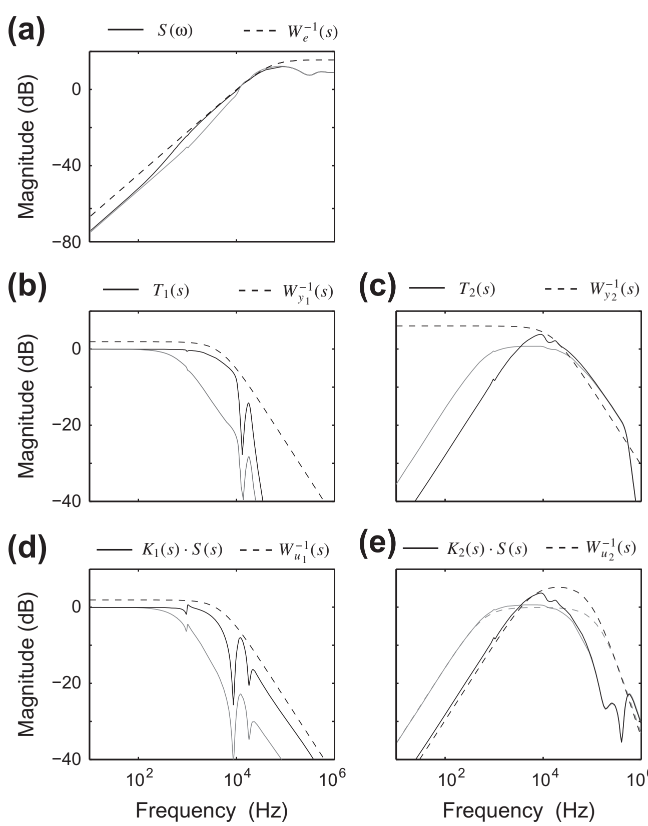

Fig. 5. Bode magnitude plots showing the results of the model-based controller synthesis. The black lines correspond to the design case aimed for the higher transition frequency (2.5 kHz) between both actuators and the gray lines correspond to the design case aimed for the lower transition frequency (500 Hz).

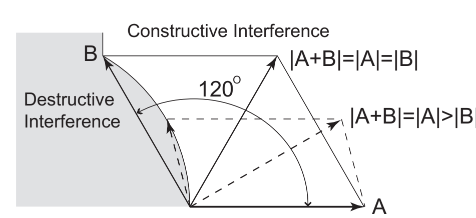

Fig. 6. Illustration of the domains of constructive and destructive interference of two actuator outputs which are represented by vectors A and B. The destructive interference (i.e. $|A + B| < max\{|A|, |B|\}$) will start at a phase difference between 90° and 120° depending on the difference in magnitude of both vectors.

would result large phase differences between both control paths, and consequently strong actuator fighting above the transition frequency. Therefore, careful choice of the transition frequency is required to minimize the chance of actuator saturation given the expected disturbance spectrum to be tracked by the system.

During typical AFM imaging experiments, the height signal to be tracked by the vertical feedback loop can be split into larger amplitude, low frequency components stemming mainly from a possible tilt of the sample and the overall shape of the sample, and smaller amplitude, high frequency components stemming from the actual nano-scale sample topography. Consequently, the height signal has a predominantly triangular shape with a base frequency equal to the line-scan frequency and an amplitude spectrum which decays inversely proportional to the frequency. In order to minimize the actuator fighting at higher frequencies, the transition frequency between both actuators is chosen such that the long-range actuator has just sufficient bandwidth to track the height variations due to the sample tilt. As for this particular system is designed for scanning speed up to about 50 Hz, a tracking bandwidth of about 500 Hz would be sufficient for the long-stroke actuator to track the height variations due to a tilt of the sample up to the 9th the harmonic. To enforce this actuator transition in the controller synthesis, weighting filter $W_{u2}^{-1}(s)$ is provided with a first order high-pass behavior with a cornering frequency of 500 Hz. The maximum gain of $W_{u2}^{-1}(s)$ is set to 1 in order to a provide higher weighting on any potential peaking of the short-range actuator controller output associated with fighting of the actuators (cf. Fig. 5e, gray lines).

In order to illustrate the importance of choosing the transition frequency between both actuators carefully, a second design case is considered in which the transition frequency is chosen such that the actuation bandwidth of the long-range actuator is pushed to its maximum 6 kHz. In this case weighting filter $W_{u2}^{-1}(s)$ is provided with a maximum gain of two and a high-pass behavior of which the cornering frequency is tuned towards the point that an actuation bandwidth of 6 kHz is achieved for the long-range actuator.

4.2. Synthesis results

The controller synthesis is performed using the Robust Control Toolbox of matlab, which in both cases results in $\gamma \leqslant 1.5$ (cf. Eq. (4)). For both design cases a disturbance rejection bandwidth is achieved of about 20 kHz (cf. Fig. 5a), and a tracking bandwidth of 40 kHz (cf. Fig. 5c). As can be seen from Fig. 5c and d, the controller and actuator outputs of the short-range actuator are slightly exceeding the magnitudes of the corresponding inverse weighting filters. However, by applying μ-analysis based on the unstructured frequency response model of Eq. (1) it is verified that for both design cases the closed loop system is robustly stable against the dynamical variations of the system, as discussed in more detail in Section 5.

The difference between both design cases is clearly revealed when comparing the contributions of both actuators. Fig. 7 shows the complementary sensitivity functions of both design cases along with the contributions of the long-range actuator (dash-dotted line) and the short-range actuator (solid). In the first design case the transition frequency between both actuators is at 500 Hz, and the peak gain of the short-range actuator is 0.7 dB (Fig. 7a), whereas in the second design case the transition frequency is at 2.5 kHz, and the peak gain of the short-range actuator is 3.8 dB (Fig. 7b). At these peaks of the short-range actuator output it can be seen that the magnitude of the complementary sensitivity function of the total system (dashed line) is significantly lower, which indicates that in those frequency regions the actuators are negating part of each others motion.

The maximum topography variation that can be tracked by the system before saturation of one of the control paths would occur can be calculated at each frequency point by:

$$d_{max}(\omega) = min\left\{\frac{r_{1_{max}}}{|S(i\omega)\cdot K_1(i\omega)|}, \frac{r_{2_{max}}}{|S(i\omega)\cdot K_2(i\omega)|}\right\}, \tag{5}$$

with $r_{1_{max}} = 5\ \mu$ m, and $r_{2_{max}} = 0.5\ \mu$ m the maximum range for the long-range and short-range actuator respectively. Fig. 8 shows the calculated maximum disturbance amplitudes $d_{max}(\omega)$ for both design cases. At lower frequency (<60 Hz) both designs utilize the full positioning range of the long-range actuator of 5 μm. Beyond the disturbance rejection bandwidth of 20 kHz both plots show a steep incline due to the fact that the controller gains at those frequencies have a steep roll-off and therefore saturation is less likely to happen. However, in the intermediate frequency region between 60 Hz and 20 kHz the short-range actuator might compromising the positioning range (cf. Fig. 8). For the design case (a) with the lower transition frequency between both actuators the positioning range starts to roll-off at about 60 Hz, while for the design case (b) with the higher transition frequency this roll off starts only at 220 Hz. On the other side, the minimum positioning range for the case (a) is 0.47 μm at a frequency of 3 kHz, while for the case (b) this is 0.33 μm at a frequency of 8 kHz. Hence, due to the actuator fighting at those frequencies, the effective positioning range is reduced by 6% and 34%, respectively, of the 0.5 μm full positioning range of the short-range actuator.

This comparison between the two designs illustrates that pushing the bandwidth of the long-range actuator higher than necessary might compromise the positioning range at higher frequencies due to the large phase differences between both control paths. Although the second design provides a large positioning range at lower frequencies, the reduction in positioning range beyond 2.5 kHz may pose higher chances of saturation of the short-range actuator while imaging rough sample surfaces at high speeds. Moreover, the larger actuator fighting in the second design case may also cause stronger excitation of the uncertain dynamics of the system in those frequency regions (cf. Fig. 3), which might degrade the reliability of the topography estimate [31]. Therefore, in the sequel of the paper only the first controller design is considered.

4.3. Controller reduction and implementation

The resulting feedback controller from the model-based controller synthesis is an 18th order one input/two outputs (SITO) system. The frequency response plots of the controller are shown in Fig. 9, showing that $K_1(s)$ has a high feedback gain predominantly at lower frequencies while the feedback gain of $K_2(s)$ is larger than $K_1(s)$ at frequencies above the transition frequency of 500 Hz, resulting in the frequency separation between both actuators. To facilitate implementation, the controller is split into two SISO controllers that are reduced to 6th and 7th order for the long-range and short-range actuator, respectively. The frequency responses of the reduced order controller are shown in Fig. 9 (dashed lines) as well, showing only small deviations from to the full-order controllers within the respective actuation bandwidths.

The feedback controllers are implemented using an *Field-Programmable Gate Array* (FPGA, Virtex-II Pro XtremeDSP Development Kit, Nallatech, Camarillo CA, USA), which runs at a sampling frequency of 65 MHz. To minimize the quantization effects on the actual pole-locations the input data is down-sampled internally by a factor of 100 for the implementation of poles and zeros for controller $K_1(s)$, and by a factor of 15 for the implementation of the poles and zeros of controller $K_2(s)$. For this implementation the poles and zeros of the respective notch filters are grouped

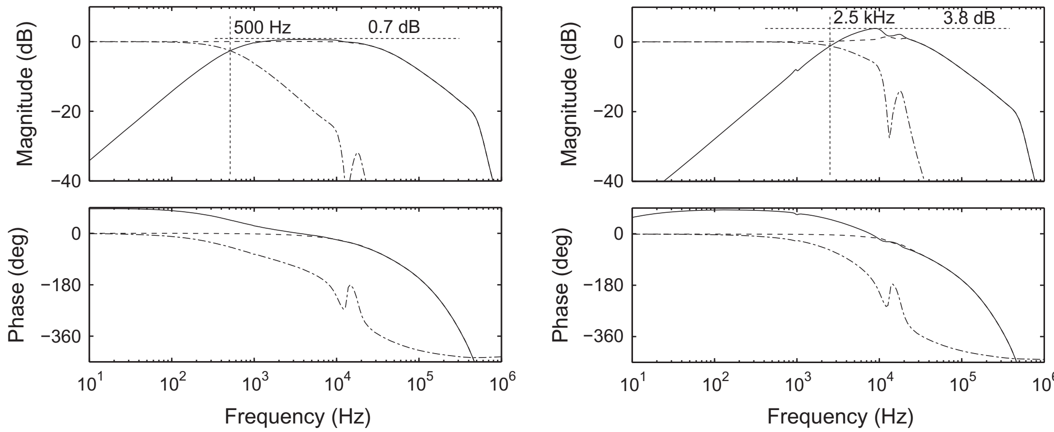

Fig. 7. Frequency response plots of the complementary sensitivity function (dashed) and the contribution of the long-range actuator (dashed-dotted), and the short-range actuator (solid) for the closed-loop controlled system with a lower transition frequency between both actuators (a), and with a higher transition frequency between both actuators (b).

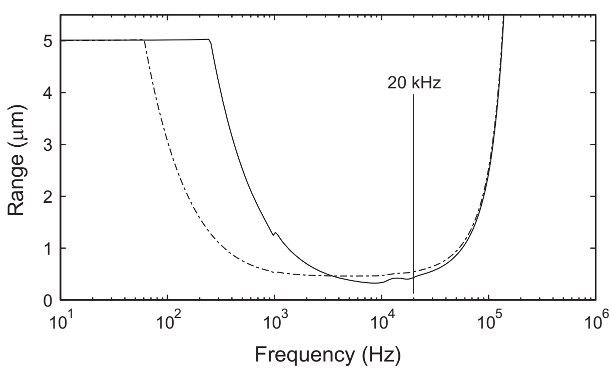

Fig. 8. The maximum allowable disturbance amplitude $d_{max}(\omega)$ before saturation of one of the control paths would occur as a function of frequency. The dashed line corresponds to the closed-loop controlled system with the lower transition frequency between both actuators, and the solid line corresponds to the system with a higher transition frequency.

in a biquad structure and discretized using a bilinear transformation via the `c2d.mat` command of `matlab`. The FPGA adds an additional 400 ns delay to the feedback loop which results in a phase lag of about 4° at the 0 dB crossing of the loop gain of the short-range actuator at 25 kHz, which is well within in the phase margin of the control system and does not influence the system performance considerably.

In order to analyze the performance of the newly designed dual actuated AFM, the disturbance rejection bandwidth is measured using the network analyzer. The results are shown in Fig. 10 (solid line) together with the modeled sensitivity function (dashed line), showing that the measured disturbance rejection bandwidth of 20 kHz closely matches with the modeled sensitivity function (Fig. 10). The achieved closed loop bandwidth is significantly higher than what can be achieved with the conventional single actuated AFM system, resulting in reduced force variations during imaging, particulary at high speeds (see Section 6).

5. Anti-windup control for dual actuated AFM

Due to the limited positioning range of the short-range actuator, saturation of this feedback path might occur at large topography variations of the sample when scanning at high speed, or large disturbances during the initialization phase of the imaging experiment. Once saturation of the short-range actuator occurs, it is important that the control-loop remains stable and the system

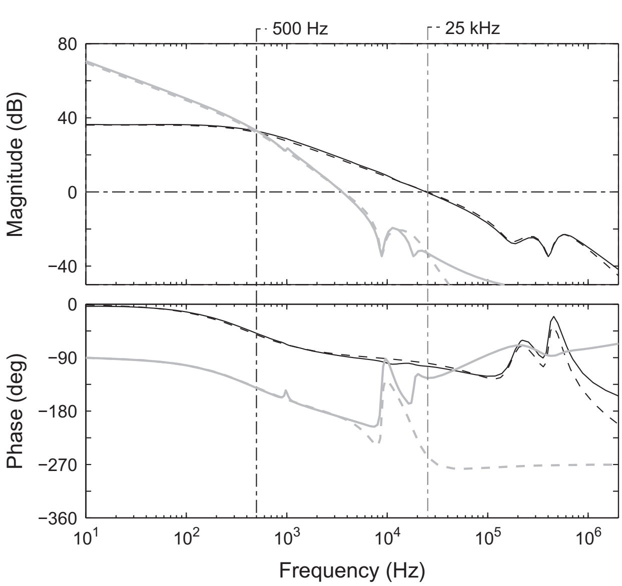

Fig. 9. Bode magnitude plots of the controllers $K_1(s)$ (gray lines) and $K_2(s)$ (black lines), full order (solid) and after order reduction (dashed).

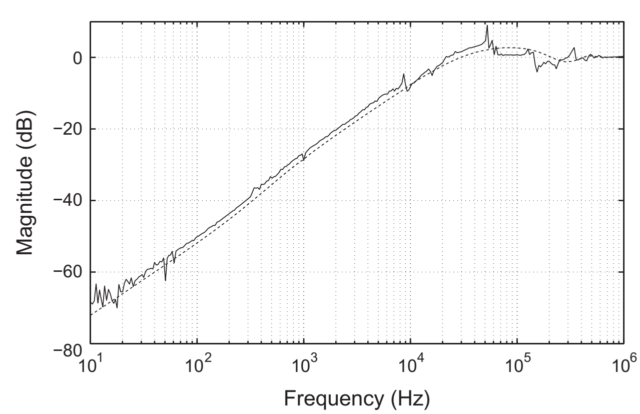

Fig. 10. Frequency response of the measured (solid) and modeled (dashed) sensitivity function.

quickly recovers to its nominal operation. This saturation problem has been addressed earlier for dual actuated HDD-system. In [32] a solution is provided to guarantee stability of dual actuated systems designed according to the decoupled Master/Slave structure. In [36] a more general solution is presented to synthesize both full-order and reduced order anti-windup controllers for dual actuated feedback systems. These methods, however, do not allow to directly take into account the dynamical uncertainty of the actuators. In this section, the robust stability of the closed-loop system is analyzed by casting the problem into a Linear–Fractional–Transformation (LFT) (see for instance [37]). Based on the obtained LFT an anti-windup controller is proposed which is shown to provide global stability of the closed-loop system with constrained controller output for the short-range actuator.

5.1. Anti-windup controller design

In the following analysis the block structure is assumed as depicted in Fig. 11a, with the designed feedback controller K, the uncertain actuator dynamics $\mathcal{G}(\omega_f, \Delta)$ (cf. Eq. (1)), and the anti-windup controller $H(s)$ that feeds back the differences between the unconstrained and the constrained control signal for the short-range actuator. The constrained controller output is given by:

$$\bar{u}_2(t) = sat(u_2(t)) = \begin{cases} sign(u_2(t)) \cdot u_{max} & , |u_2(t)| > u_{max} \\ u_2(t) & , |u_2(t)| \leqslant u_{max} \end{cases}, \tag{6}$$

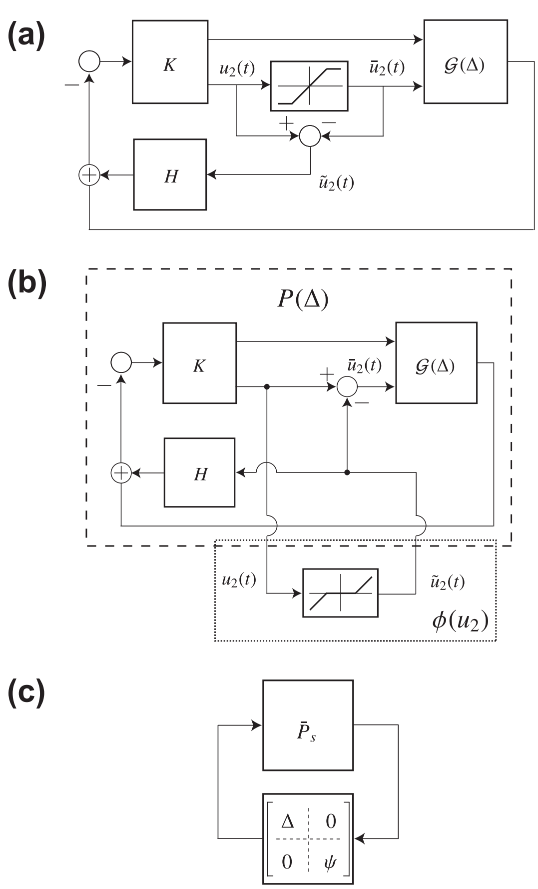

Fig. 11. (a) Block diagram of the constrained dual actuated system with anti-windup compensator H. (b) Alternative system representation where the saturation block is replaced by a dead-zone non-linearity. (c) Linear fractional transformation of the normalized system $\bar{P}_s$ interconnected with the dynamical uncertainty Δ and the normalized output non-linearity ψ.

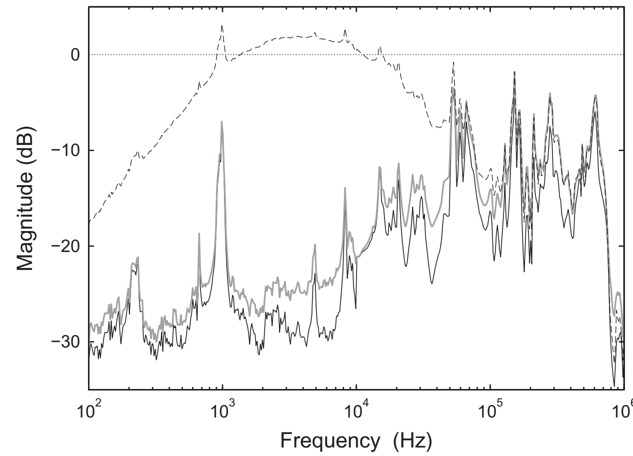

Fig. 12. Structured Singular Value (SSV) plot for the unconstrained system, disregarding the actuator saturation (black, solid line), for the constrained system without anti-windup controller $H(s) = 0$ (black, dashed line), and for the constrained system with a static anti-windup controller $H = G_2(s = 0)$ (gray solid line).

Fig. 11b shows an alternative representation of this saturation problem where the saturation non-linearity is replaced by a dead-zone non-linearity [38], of which the output equals the difference between the unconstrained and constrained controller output:

$$\tilde{u}_2(t) = \phi(u_2(t)) = u_2(t) - \bar{u}_2(t) = u_2(t) - sat(u_2(t)). \tag{7}$$

The dead-zone non-linearity $\phi(u_2(t))$ can be seen as a non-linear gain belonging to the sector $[0, 1]$, and is interconnected with the transfer from $\tilde{u}_2(t)$ towards the $u_2(t)$ denoted $P(\omega_f, \Delta)$ (cf. Fig. 11b):

$$P(\omega_f, \Delta) = \frac{K_2(i\omega_f)(\mathcal{G}_2(\omega_f, \Delta_2) - H(i\omega_f))}{1 + K(i\omega_f) \cdot \mathcal{G}(\omega_f, \Delta)}, \tag{8}$$

Following the 'loop transformation' procedure as described in [38], this interconnected system can be normalized by defining $[\psi(u_2) = 2 \cdot \phi(u_2) - I]$ which belongs to the sector $[-1, 1]$, and

$$P_s(\omega_f, \Delta) = \frac{0.5 \cdot P(\omega_f, \Delta)}{1 + 0.5 \cdot P(\omega_f, \Delta)}. \tag{9}$$

By also pulling out the dynamical uncertainty Δ the system can be represented by the LFT of Fig. 11c. Based on this LFT the robustness of the system can be analyzed for both the dynamical uncertainty Δ as well as the non-linearity posed by the actuator

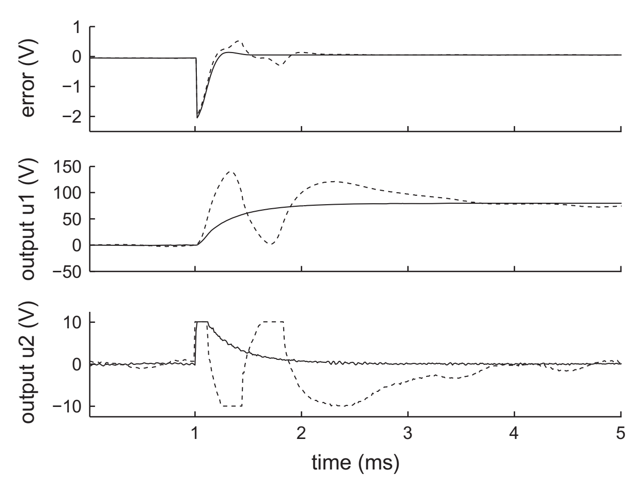

Fig. 13. Measured system response to a reference step input of about 1 μm which results in saturation of the control output for the short-range actuator ($u_2(t)$), shown with (solid) and without (dashed) static anti-windup compensation.

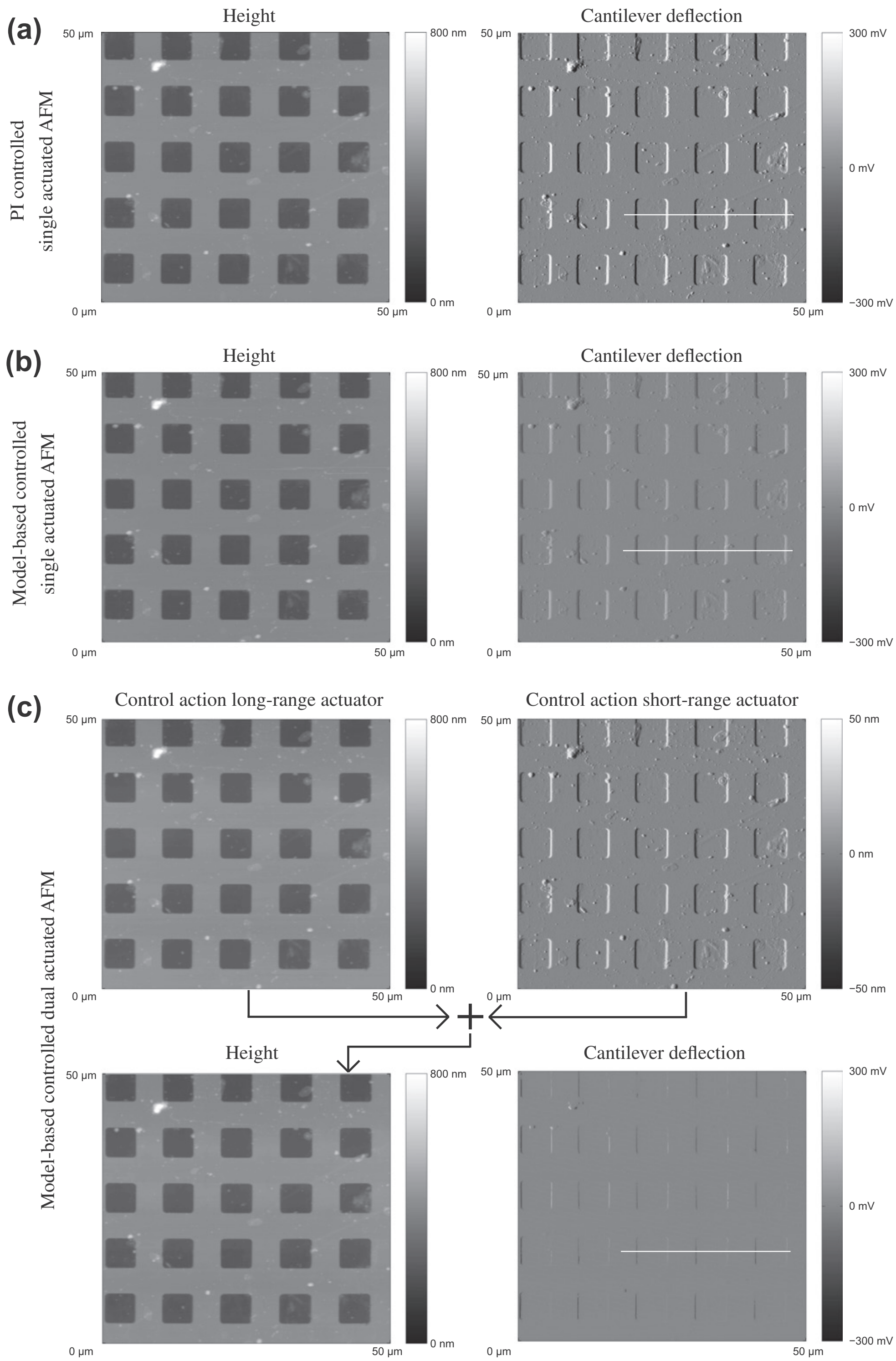

Fig. 14. AFM images of the same area of a silica calibration grating (10 μm pitch, 200 nm step height), obtained with the single actuated PI controlled AFM (a), the model-based controlled single actuated AFM (b), and the newly proposed model-based dual actuated AFM. (c) The images recorded with the dual actuated AFM show a significant reduction of cantilever deflection, and thus reduced force variations between the measurement tip and the sample as compared the both single actuated AFM's. The white line in the deflection images indicates the cross-sections shown in Fig. 15.

saturation by applying μ-analysis [37]. Global asymptotic stability of the system is assured when the calculated Structured Singular Value (SSV) $\mu(\omega)$ of the system is lower than one for all frequencies ($\mu(\omega) \leqslant 1, \forall\omega \in \Omega_f$), which can be calculated with the `mussv.m`-function of the robust control toolbox of `matlab`. The black solid line of Fig. 12 shows the calculated SSV-plot for the system when only regarding the dynamical uncertainty Δ, and the actuator saturation is neglected. As for this case SSV-plot is below the 0 dB line for all frequencies, it can be concluded that the feedback controller K designed in the previous section indeed provides robust stability of the unconstrained system. The dashed black line of Fig. 12, however, shows the SSV-plot when taking the actuator saturation into account, but without implementation of an anti-windup controller (i.e. $H(s) = 0$). As for some frequencies this

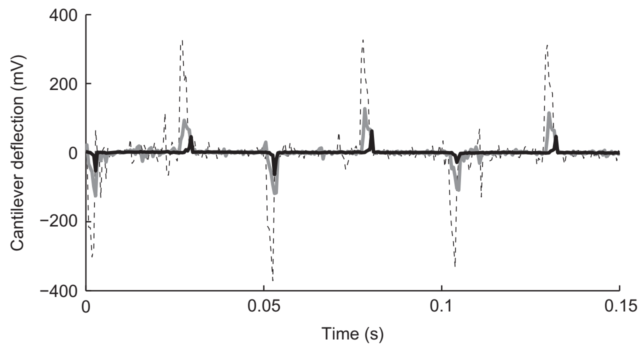

Fig. 15. Cross-section of the cantilever deflection signal for the PI-controlled single actuated AFM (dashed, black), the model-based single actuated AFM (solid, gray), and the dual actuated AFM (solid, black).

SSV-plot exceeds the 0 dB line, global asymptotic stability of the system without anti-windup controller cannot be guaranteed. Experimental results (data not shown) demonstrate that the constrained system is indeed going unstable when tracking step-like feature slightly larger than 1 μm.

A well known design choice for the anti-windup controller is the Internal Model Control (IMC) [39] method in which a model of the nominal actuator dynamics is taken as the anti-windup filter $H(s)$. During saturation the actuator model mimics the behavior of the short-range actuator such that the rest of the loop is not affected by the saturation. If the actuator model matches the true nominal dynamical behavior of the actuator (i.e. $H(\omega) = \mathcal{G}_2(\omega)$) the IMC method fully compensates for the stability issues associated with the actuator saturation, and the calculated SSV-plot overlaps that of the unconstrained system (black solid line of Fig. 12). For practical implementation, however, the anti-windup controller should be taken as a finite order structured dynamical model. An important design consideration hereby is that although using a high order model for the anti-windup controller which closely matches the true dynamics of the constrained actuator may provide the best robust stability of the system, the transient behavior of the system may be compromised by such high order anti-windup controller. This results from the energy build up in the dynamical model $H(s)$ during the saturation phase, which can cause oscillations as soon as the system recovers from saturation, particulary when the actuator model contains weakly damped resonant modes [39]. Moreover, the use of a high order anti-windup controller might not always be implementable, or may consume a lot of computational power of the real-time hardware. However, as for this particular system the feedback bandwidth of 40 kHz is well below the first fundamental resonance frequency of the short-range actuator at 150 kHz (cf. Fig. 2), it would adequate to consider the short-range actuator as a static gain in the design of the anti-windup controller: $H = G_2(s = 0)$. The use of such a static anti-windup controller requires minimal computation power, and problems with poor transient performance associated with energy buildup in H does not occur. Fig. 12 (gray solid lines) shows the calculated SSV-plot for the system with the static anti-windup controller, showing that although the robust stability is slightly less then in case of the unconstrained system, the SSV-plot is lower that 0 dB over all frequencies, and therefore global asymptotic stability of all systems within the uncertainty set of Eq. (1) is guaranteed.

5.2. Experimental validation of the anti-windup controller

The static anti-windup controller is implemented on the FPGA along with the design feedback controller. Fig. 13 shows the response of the experimental system with and without the proposed static anti-windup controller to a step at the input with a height of 1 μm. In the case without the anti-windup controller (dashed lines) large oscillations are visible in both the error signal as well as in the control signals for the long- and short-range actuator, while the case with the static anti-windup controller (solid lines) shows a smooth recovery from the actuator saturation. Experiments show that the system without anti-windup controller goes unstable for step inputs slightly larger than 1 μm, while the system with the static anti-windup controller is stable over the entire working range. Hence, the static anti-windup controller is a simple but essential extension of the dual-actuated control system, which guarantees stability of the system under all imaging conditions.

6. Imaging results

In order to demonstrate the benefits of dual actuation to control the tip-sample force in AFM, three sets of AFM images are captured with different feedback controllers: (i) the single actuated system with a conventional PI controller given by the commercially available system as a benchmark, (ii) the single actuated system with a model-based feedback controller, and (iii) the newly designed dual actuated system with model-based feedback controller with the disturbance rejection bandwidth of 20 kHz (cf. Fig. 10) and static anti-windup controller. The PI controlled system is the well-tuned standard controller of the `Multimode V` system, and achieves a closed-loop disturbance rejection bandwidth of 400 Hz. The model-based controller for the single-actuated system is designed according to the procedure as described in [19], and achieves a disturbance rejection bandwidth of about 1 kHz. Fig. 14a–c shows AFM-images of the same area of a calibration grating with step height of 200 nm, and 10 μm pitch obtained at a scan-speed of two lines per second with the prototype AFM system controlled by the three different controllers. For each experiment both the topography measurement as well as the measured cantilever deflection are shown. Fig. 14 clearly shows that the cantilever deflection of the model-based single actuated system is significantly lower than the cantilever deflection obtained with the single actuated PI controlled AFM as a results of the higher control bandwidth. For the dual actuated AFM (Fig. 14c) the cantilever deflection is even further reduced, only showing small variation around the sharp edges of the calibration grating. Fig. 15 shows cross-sections of the different cantilever deflection images as marked by the white lines in the deflection images in Fig. 14, confirming that with the dual actuated AFM the cantilever deflection is significantly reduced as compared to both single actuated AFMs. Fig. 14c also shows images of the feedback controller outputs for both actuators in the dual actuated case, demonstrating that the larger amplitude, low frequency topography variations are tracked by the long-range actuator, and that the short-range actuator only tracks the high frequency topography variations that otherwise would occur in the cantilever deflection image (cf. Fig. 14a). Furthermore, the topography measurement is obtained by summing both feedback controller outputs.

As demonstrated by Fig. 14 the higher control bandwidth in the dual actuated case, results in less force variations between the tip and the sample and thus more gentle probing of the sample, which is particularly important when imaging fragile biological samples. The 20 times higher control bandwidth would allow 20 times faster imaging as compared to the model-based controlled, single actuated system with equally large force variation between the tip and the sample.

7. Conclusions

This contribution presents the design and implementation of a model-based feedback controller for a dual actuated AFM (prototype). In order to identify the dynamical behavior of the system for varying imaging conditions, the frequency responses of both actuators are measured with various measurement probes and sample holders. Based on the measured frequency responses, dynamical models for both actuators including an uncertainty description are derived, and the maximum allowable actuation bandwidths are determined. Based on these models, a model-based feedback controller is designed to control the tip-sample interaction force via both actuators simultaneously. Special emphasis is given on providing sufficient attenuation of the dynamic uncertainty to achieve robust stability under all working conditions, while also preventing strong destructive interference between both actuators. In order to ensure stability of the closed-loop under possible saturation of the short-range actuator, an anti-windup controller is presented which is shown to provide stability of the system under all imaging conditions. The designed controller is implemented on the prototype system by utilizing an *FPGA*, achieving a closed-loop disturbance rejection of about 20 kHz, which is about 20 times higher than the model-based controlled single actuated system.

AFM-images are obtained with the newly designed dual-actuated AFM system in order demonstrate the significant reduction in residual control error, given by the cantilever deflection, as compared to the single actuated AFM that is controlled either by a PI-controller or a model-based controller. The reduced force variations between the tip and the sample results in a more gentle probing of the sample surface, and allows for faster imaging which is especially important when imaging fragile biological samples.

In this research it is shown that the dynamical uncertainty of the system under various imaging condition may pose a limitation on the achievable closed-loop bandwidth for controlling the tip-sample force. Therefore, reducing the uncertainty of the system dynamics for all potential imaging conditions may be an important additional design criterion for future high-speed AFMs, in order to push the closed-loop bandwidth even higher without jeopardizing system stability. Moreover, the reducing the dynamic uncertainty of the system may improve accuracy of the sample topography estimation, as discussed in [31]. The design methodology proposed in this research may be used for analysis as well as controller synthesis for the next generation of high-speed AFMs.

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